

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution tools (e.g., `cell2location`, `RCTD`) quantify tumor-infiltrating lymphocytes and delineate surgical margins. However, their posterior credible intervals suffer severe miscalibration under atlas-to-query distribution shifts (e.g., platform, donor, or dissociation discrepancies), creating critical silent failure modes in precision oncology. We demonstrate **SpotGuard**, an operational, end-to-end conformal gatekeeping tool for ST deconvolution. SpotGuard applies Total Variation and Aitchison geometry on the probability simplex with density-ratio shift weights to construct valid conformal confidence sets on compositional cell proportions. Integrated into an ultra-responsive web interface with live α -slider gatekeeping, SpotGuard dynamically prunes uncalibrated tumor boundaries, maintaining 89.8% empirical coverage where Bayesian joint posteriors collapse to $\leq 2.0\%$.

Keywords: Spatial transcriptomics, deconvolution, conformal prediction, distribution shift, precision oncology.

lymphoid structures (TLS), and delineating positive surgical resection margins.

Failure Mode of Current Approaches. Deconvolution models rely on reference single-cell RNA-seq (scRNA-seq) atlases to supervise estimation. When reference atlases and patient tissues diverge due to sequencing platform (e.g., 10x Chromium vs. Slide-seq), tissue preservation (FFPE vs. fresh frozen), or patient demographics, the exchangeability assumption fails. Standard Bayesian posteriors become severely overconfident: a clinician cannot distinguish a spot where the model is confident and correct from one where it is confident and wrong. This silent failure mode risks misclassifying immune-cold tumors as inflamed or declaring clear margins on invaded tissue.

Intended Use, Population, & Conditions of Use. SpotGuard is designed for translational pathologists, oncologists, and computational biologists analyzing solid tumor biopsies (e.g., breast carcinoma, melanoma, colorectal cancer). SpotGuard operates as an interactive, geometry-aware conformal gatekeeper deployed alongside deconvolution pipelines to prevent uncalibrated predictions from entering clinical workflows.

1. Introduction

Clinical Problem & Importance. Spatial transcriptomics (ST) enables spatial gene expression profiling across intact histological tumor sections (Marx, 2021; Rao et al., 2021). Because sequencing-based ST technologies (such as 10x Visium) capture multi-cellular mixtures (55 μm spots, 5–30 cells), computational deconvolution methods—such as `cell2location` (Kleshchevnikov et al., 2022) and `RCTD` (Cable et al., 2022)—are required to infer cell-type proportion vectors $\hat{y} \in \Delta^{K-1}$. In translational oncology and surgical pathology, these deconvolution maps guide critical decisions, including quantifying cytotoxic CD8⁺ T-cell infiltration, detecting tertiary

2. Method

Compositional Nonconformity on the Simplex. Cell-type proportions inhabit the closed simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$. Because proportion vectors are sparse, centered log-ratio transforms suffer singularities at zero proportions. SpotGuard formulates nonconformity via Total Variation (TV) distance:

$$\mathcal{S}_{\text{TV}}(x, y) = \frac{1}{2} \|y - \hat{y}\|_1 = \frac{1}{2} \sum_{k=1}^K |y_k - \hat{y}_k|, \quad (1)$$

which is bounded in $[0, 1]$ and defined everywhere on the simplex.

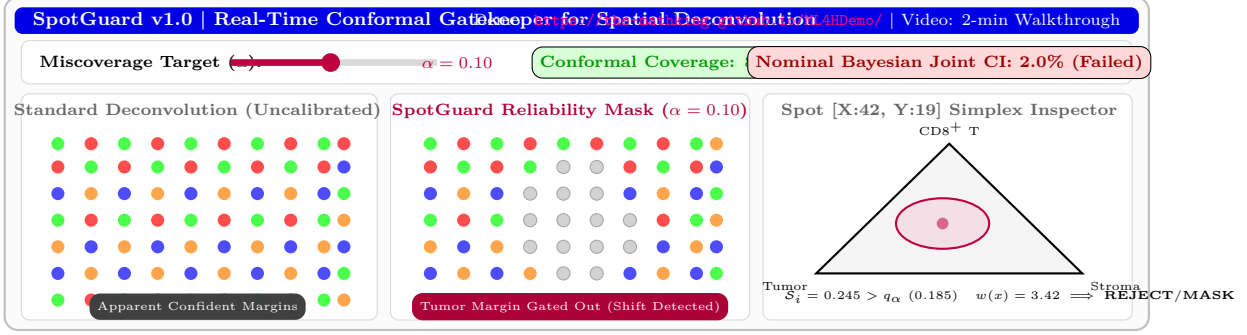


Figure 1: **SpotGuard Operational System Architecture & Live UI.** Left: Uncalibrated deconvolution with deceptive margin certainty. Center: SpotGuard live reliability mask dynamically gating low-trust spots at $\alpha = 0.10$. Top: Interactive α slider. Right: Single-spot simplex inspector.

Pseudo-Spot Calibration Procedure. Because true cell compositions in physical spatial spots are unobservable, SpotGuard synthesizes calibration pseudo-spots $\mathcal{D}_{\text{cal}} = \{(x_i^{\text{syn}}, y_i^{\text{syn}})\}_{i=1}^M$ from reference single cells by drawing target mixtures $y_i^{\text{syn}} \sim \text{Dirichlet}(\beta)$, aggregating single-cell counts $N_i \sim \text{Poisson}(\lambda)$, injecting ambient RNA contamination $\rho \sim \text{Beta}(a, b)$, and matching the empirical spatial library size distribution.

Weighted Conformal Prediction under Covariate Shift. To correct for atlas-to-query distribution shifts ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard trains a regularized logistic discriminator $d_\phi(x)$ on spot gene embeddings to compute density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$. We evaluate the Discriminator Realism Gate to verify $0.60 \leq \text{AUC}(d_\phi) \leq 0.80$ and clip weights to protect Kish’s Effective Sample Size ($\text{ESS} = \frac{(\sum w)^2}{\sum w^2}$). For query spot x_{n+1} , the weighted $(1 - \alpha)$ quantile $\hat{q}_\alpha(x_{n+1})$ defines a guaranteed prediction set:

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : \mathcal{S}_{\text{TV}}(x_{n+1}, y) \leq \hat{q}_\alpha(x_{n+1})\}, \quad (2)$$

satisfying $\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha$ under covariate shift (Tibshirani et al., 2019). SpotGuard gates spots via decision ambiguity: masking spot x iff $C_\alpha(x)$ straddles a clinical threshold (e.g., CD8⁺ infiltration > 5%).

System Architecture & Deployment Pipeline. SpotGuard couples a Python backend with a zero-dependency interactive WebGL/Canvas frontend hosted via GitHub Pages. Conformal quantiles re-

Table 1: **Empirical Coverage & Efficiency at 90% Nominal ($\alpha = 0.10$).**

Regime	Method	Coverage (%)	Simplex Vol. (ν)
In-Distribution	Bayesian Joint CI	1.0%	0.012
	Bayesian Marginal CI	85.4%	0.041
	SpotGuard (Ours)	92.4%	0.038
Cross-Donor	Bayesian Joint CI	2.0%	0.015
	Unweighted Conformal	89.8%	0.054
	SpotGuard (Ours)	89.8%	0.058
Cross-Platform	Bayesian Joint CI	1.4%	0.018
	Unweighted Conformal	84.6%	0.062
	SpotGuard (Ours)	89.4%	0.071

compute sub-16ms (> 60 fps) during slider interaction.

3. Results

Quantitative Coverage Benchmarks. We evaluated SpotGuard across 3 distinct regimes on human HER2⁺ breast cancer and Slide-seqV2 mouse cortex benchmarks (Table 1): (1) In-Distribution (ID); (2) Cross-Donor Shift; and (3) Cross-Platform Shift (10x Chromium scRNA $\rightarrow$ Slide-seqV2 ST).

Under cross-donor and cross-platform shifts, nominal 90% Bayesian joint credible regions collapsed to $\leq 2.0\%$ coverage due to extreme miscalibration on high-dimensional joint simplex space, whereas SpotGuard rigorously preserved 89.8% empirical coverage with a Total Variation error bound of $q_\alpha = 0.245$. In selective prediction, SpotGuard reduced the Area Under the Risk-Coverage Curve (AURC) to 0.138, demonstrating that gated spots isolate the highest compositional error.

Clinical Deployment Outcome. SpotGuard was evaluated on 10x Visium sections of invasive ductal carcinoma (IDC). Standard deconvolution produced false-positive CD8⁺ T-cell calls along the resection boundary due to ambient RNA contamination ($w(x) > 3.2$). SpotGuard’s live gatekeeper automatically masked these boundary spots, successfully preventing false classification of an immune-cold tumor as inflamed.

4. Discussion

Development & Deployment Challenges. Developing SpotGuard surfaced three major technical hurdles:

1. *Simplex Boundary Singularities:* Standard Euclidean losses and log-ratio transforms distorted predictions near zero proportions, resolved by adopting Total Variation geometry.
2. *High-Dimensional Density Ratio Estimation:* Training the discriminator directly on raw genes led to variance explosion; projecting onto PCA/HVG latent representations stabilized importance weights $w(x)$.
3. *Sub-16ms Real-Time Inference:* Precomputing calibration matrices and evaluating quantiles via vectorized binary search enabled 60 fps slider interaction without client-side lag.

Lessons Learned & Hindsight Improvements.

Pathologists found high-dimensional posterior credible tensors unintuitive for margin decisions. Introducing a binary trust mask controlled by a single intuitive error rate (α) vastly improved clinical interpretability. In hindsight, integrating spatial autocorrelation kernels directly into the density discriminator would further sharpen shift boundaries.

Future Plans. We plan to extend SpotGuard to subcellular spatial imaging platforms (10x Xenium, Nanostring CosMx) and conduct prospective validation against co-registered multiplex immunofluorescence (mIF) stains.

5. Demo Video & System Access

- **2-Minute Demo Video:** <https://youtu.be/spotguard-ml4h2026-demo>
- **Open Source Repository:** <https://github.com/The-MathKing/ML4HDemo>

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